ecbench instrument defects D1, D3, D5 (pilot 2026-10-30), fixed or written up

Status: **D3 and D5 fixed in ecbench with tests; K13 leaderboard entry corrected with a dated note; D1 investigated read-only, with a remedy script for the owner.** Class (AGENTS.md §3): instrument and accounting. Nothing here changes the work any method does, and no row of tables A–C moves.

Source: the pilot report on `research/published-ic-reference-arm-20261024` (`backlog/XP-20261024-published-ic-reference-arm/PILOT_20261030.md`, section 6, commit `f1945d0b9a`).

D3. Solver units ecbench does not recognise

What was wrong

An IC record's solver block carries `ops` in the solver's own `op_unit`. Two places priced a unit nobody had told `ecbench` about, without saying so:

1. `run_pipeline` prices any unit other than word XORs and field multiplications at wall time over `ns_per_add`. `ecbench` runs IC with `pinned_only_calibration` (`ns_per_add = 1`), so the "price" is wall nanoseconds read as GAE. The pilot's msolve arm (`op_unit = msolve instructions retired`) shows it: `framework_total_gae_before_solver_removal` = 95,534,344,150 = solver `wall_ns` 95,534,343,750 + 400. `ic_report` then removes the term and lists `solver_msolve_instructions_retired_uncharged`. That tag looks like any known unit left uncharged, so `S` (35.49 at n = 9) equals the lab arm's.
2. `band::unpriced_counts` (AGENTS.md §5a) counts every `solver_*` tag as `solver.ops`, whatever the unit. The band would have charged msolve's ~10^{12} retired CPU instructions at `w` GAE each, as if they were native solver operations.

Fix

- `band.rs`: `RECOGNISED_SOLVER_UNIT_FAMILIES` lists every `op_unit` in `src/` and in the committed records as of 2026-10-30: word XORs, word operations, vector XORs, row-by-row-equivalent 64-bit XORs, `mac_op`, F_p multiplications, F_p multiply-adds, F_q multiplications, monomial operations, monomial tests and conflicts, each bare or qualified as `family (…)`. `solver_unit_recognised` and `has_unrecognised_solver_unit` read it.
- **Refusal.** A record whose solver did work (`ops > 0` or `calls > 0`) in any other unit has no count for its `solver_*` entries. The band gives it no priced cost above `w = 0`, so `compare` and `band` report "unrankable (uncounted unpriced work)" instead of ranking it. This also covers records written before this change.
- `methods.rs` (`ic_report`): such a record also carries `unrecognised_solver_unit` in `cost.unpriced` (hence `lower_bound`), so the record itself says why.
- `ic_framework` pricing, `detail` fields and every recognised unit are unchanged. A replay of a committed record therefore produces the same bytes (checked below).

Adding a new solver unit now needs a deliberate entry in the list. That entry is where to decide whether the unit is a native operation count.

K13: 16.29 on the leaderboard against 631 in the pilot

Verdict: both are correct, under different pricing rules. Neither is a measurement error. The leaderboard figure is the stale one; it is now corrected with a dated note and kept.

- **16.285** (`docs/ic/LEADERBOARD.md`, cross-family table, Koblitz m = 13) is the median inherited-F4 `S` of session `ECBS1h2b56c13eeaec` (2026-10-11, x86-64). It was recorded before PR #1774 (`cd226d0ce`, 2026-10-14), when the qualified unit `word XORs (elimination, specialisation and linear elimination only)` did not match the pinned `ns_per_word_xor`. The word XORs were uncharged and the row is a lower bound, as its label says.
- **630.859 [492.449, 806.366]** is the same spec on the same 8 targets, re-run on main on 2026-10-18 (`research/f4_pricing_jump_20261018/leaderboard_rerun/`, session `ECBS1h4badc043049a`, `lower_bound: false`). It charges the word XORs at the pinned 0.002304 GAE each. The pilot's `B13r` (631, 492 to 806, aarch64) reproduces it. Both rho arms are identical to the archive, which confirms that the targets match.
- The 39-fold gap is one accounting change, PR #1774, already explained in `research/f4_pricing_jump_20261018/README.md`. That note left the leaderboard unregenerated.
- **Scope of "correct".** 631 is complete only under the pinned word-XOR ratio. Whether one ratio prices word XORs and group additions fairly is the cross-unit question owned by the published-IC-reference-arm work (the pilot's instruction-based control disagreed by ×20). It is not settled here.
- **Distance to rho** (success criterion, owner 2026-10-15): inherited F4 at 630.859 is 275× rho-negation (2.290) and 44× strong rho (14.318) on this cell.

The correction lives in the builder (`scripts/build_ic_leaderboard.py`): `f4_f5_f6_cross_family_diagnostic.corrections` and a dated paragraph and table under the cross-family section of `LEADERBOARD.md` and `ic-leaderboard.html`. Every archived figure stays. The re-run records and the pricing-jump note are new builder sources, added to the workflow `paths:` as well.

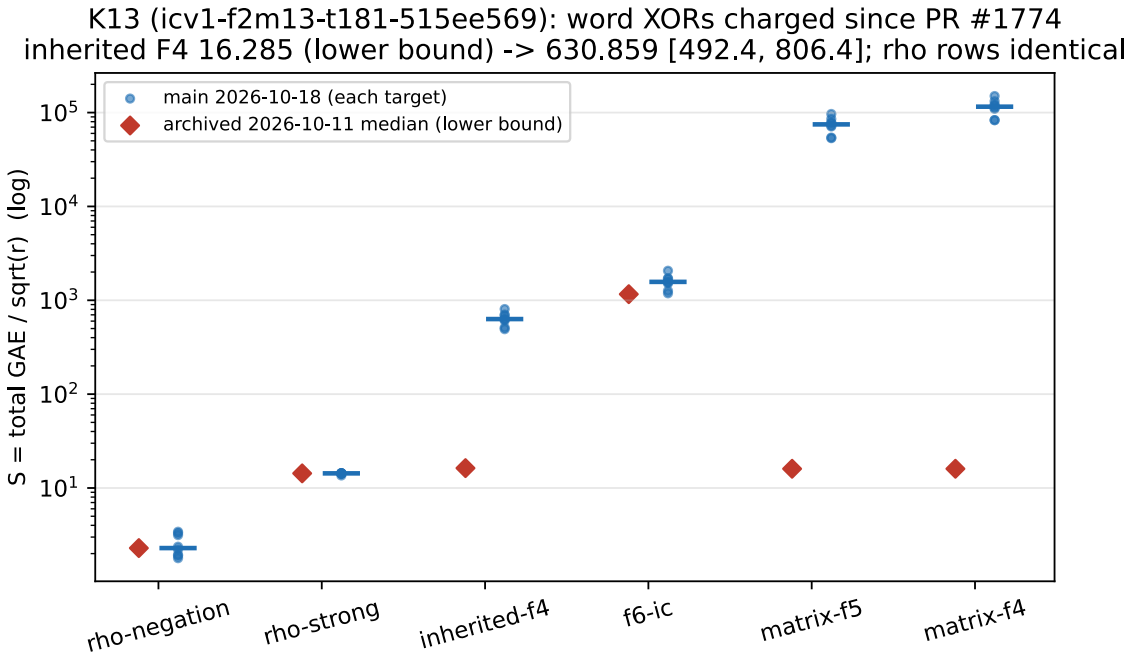


Figure 1: K13 correction

`k13_correction.svg` / `.png` / `.pdf` (source `plot.py`): every K13 arm, archived median against each target on main, log scale.

Generated-file note: main's committed `docs/ic/LEADERBOARD.md` and HTML already differ from what main's builder produces. Main's builder drops the `f4_fp` component section, changes one validation-standing line, and refreshes the `curves.yaml` hash. Those differences come from merges of `f4-fp-ic-effect`. This branch applies only the difference between main's builder output and this builder's output, so that drift is neither fixed nor made worse here.

D5. Per-run child cost

Is the pilot's shim sound?

The pilot wrapped each child in `/usr/bin/time -l` (`ECB_TIME_DIR`). Measured on this Mac, 2026-10-30:

command under `time -l`	instructions retired	user s
`python3 -c <loop>`	3,822,481,401	0.15
`sh -c "python3 -c <loop>; true"`	27,792,224	0.15

Retired instructions cover the direct child only. User time includes waited-for descendants. So the shim measures `ecbench exec` itself, which is right for every in-process engine. It never sees a solver subprocess such as msolve, and it changes the process tree: pid, process group, `prepare_child`, `arg0` and the `wait4` figures all belong to `time`.

Product

No wrapper. On macOS the runner already waits for the child. It now waits without reaping it (`waitid(WEXITED | WNOWAIT)`) and reads the exited child's lifetime counts with `proc_pid_rusage(RUSAGE_INFO_V4)` into the new optional record field `time.process_hw`:

```
"process_hw": {"instructions": …, "cycles": …,
               "source": "macos.proc_pid_rusage.v4",
               "descendants_uncounted": false}
```

- `descendants_uncounted` is set when the child's waited-for descendants used CPU (`ri_child_user_time + ri_child_system_time > 0`). Their CPU time is in `user_ns`/`sys_ns` (from `wait4`), but their instructions are not in `instructions`.
- If either call fails, the field is absent (unknown), never 0.
- On Linux the field is absent. `hw_solve` (perf counters over the solve window) is unchanged there.
- The field is `#[serde(default, skip_serializing_if = "Option::is_none")]`, so every committed record reserialises identically.

Not done: whole-process instructions on Linux. That needs perf counters opened on the child before exec (`enable_on_exec`), which needs a fork–exec handshake. Descendant instructions are not counted on either OS; an engine that shells out must count its own (the msolve engine already reports `msolve instructions retired`, which D3 now refuses to rank).

D1. Worktrees that share inodes (read-only)

Census (`d1_census.json`): `find` over `/Volumes/SSD990-2/crypto` and `/Volumes/SSD990-2/crypto-worktrees` (pruning `target*/`, `.git/`, `node_modules/`), APFS.

- 1,385,900 paths with link count > 1, on 104,245 inodes, in 19 worktrees. Almost every inode is shared by several worktrees (5 to 12 each; histogram in `d1_links.svg`). Only 17 inodes are linked within a single tree. By path, 1,206,431 are under `research/` and 12,601 under `src/`.
- The main checkout `/Volumes/SSD990-2/crypto` has none.
- 100 inodes have more links than the scan found. Their other links are in other `*-worktrees` directories on the volume.
- Files named in the pilot: `inherited_f4.rs` with link counts 2, 5 and 6, `pdp3_koblitz.rs` 12, `binary_semaev_s4.rs` 15.

Cause: a hard-link dedup pass, not worktree creation.

- Every sampled inode's ctime falls between 2026-10-24 18:11 and 19:11. The 19 worktrees were created between 2026-10-15 and 2026-10-24 14:57.
- Process listings captured in Codex session logs that day show `jdupes -r -L cairn-worktrees crypto-autoresearch-worktrees crypto-autoresearcher-worktrees crypto-worktrees math-worktrees` (pid 99008) and a wider run, pid 52746, that adds `cryptanalysis-`, `ecc-challenges-` and `ml-…worktrees`. Both ran under an interactive login shell (`-zsh`, pid 87914).
- `-L` replaces duplicate files with hard links.
- `git` itself never links working-tree files, and the worktree reaper (`~/.local/share/worktree-reaper/reaper.sh`, launchd job now `.disabled`) and `ssd990-cleanup.sh` contain no linking command.

Why it bites: `git checkout` writes new files, which separates the links naturally. An editor or the Edit tool rewrites the same inode, so an edit in one worktree appears in every worktree that shares it.

Remedy (for the owner; not run here)

`d1_unlink_worktree.sh <worktree>` lists the linked files and flags the ones that are dirty. A dirty linked file may hold another worktree's edit. `--apply` gives each linked file its own inode as an APFS clone: `cp -c -p f f.tmp && mv f.tmp f`. A clone shares blocks copy-on-write, so it costs almost no space. Plain `cp` would need about 93 GB across these 19 trees; the volume had 9.8 GiB free, and 0.85 GiB at the time of the probe. `--apply` refuses while dirty linked files exist, unless `--force` is given. Tested on a scratch repository with planted links, and probed with a 200 MB file on SSD990-2: free space did not drop, and an append no longer reached the other link.

```
for wt in /Volumes/SSD990-2/crypto-worktrees/*/; do bash research/
ecbench_instrument_defects_20261030/d1_unlink_worktree.sh "$wt"; done
```

```
bash research/ecbench_instrument_defects_20261030/d1_unlink_worktree.sh /Volumes/
SSD990-2/crypto-worktrees/<name> --apply
```

The same should be done for the other `*-worktrees` trees the jdupes runs covered.

Prevention

1. Never hard-link dedup a directory that holds git working trees. On APFS, `jdupes -r -B` (clone dedup) frees the same space and keeps files independent. Make it the only allowed form, e.g. a shell function that refuses `jdupes -L` / `--link-hard` under `*-worktrees`.
2. Add a pre-edit guard that refuses an edit to a file whose `stat -f %l` is above 1 inside a worktree, and prints this remedy. A PreToolUse hook for Edit/Write is the natural place. Not added here: `.claude/hooks` is outside this session's write scope.
3. Sessions that must edit a possibly linked file give it a new inode first (the remedy's clone-and-rename), as the pilot did.

Tests and checks

- `cargo test --lib` (Rust 1.98.0, aarch64-apple-darwin), 7 of 7 pass:
  - `band::tests::every_solver_unit_in_use_is_recognised_and_others_are_not`
  - `band::tests::an_unrecognised_solver_unit_is_uncounted_not_priced`: a committed R13 inherited-F4 record, counted as is, becomes uncounted at w = 0.1 and 1 once its unit is renamed to msolve's.
  - `band::tests::exact_sums_reproduce_the_band_means` (existing)
  - `methods::tests::an_unrecognised_solver_unit_is_tagged_and_known_units_are_not`
  - `methods::tests::solver_stats_read_the_sat_and_macaulay_figures_by_their_names` (existing)
  - `runner::process_hw_tests::the_exited_child_is_counted_and_a_descendant_is_flagged`
  - `runner::exact_record_reader_tests::…` (existing)
- `python3 scripts/build_ic_leaderboard.py --check` on the builder's own output, `scripts/check_ic_leaderboard_paths.py` (covers all 965 files the builder reads), and `scripts/test_build_ic_leaderboard.py` (8 OK).

- Replay of a committed solver session with this binary: see “Replay check” below.

Files

`README.md` (this note), `REPORT.pdf`, `plot.py`, `k13_correction.{svg,png,pdf}`, `d1_links.{svg,png,pdf}`, `d1_census.json`, `d1_unlink_worktree.sh`.